$$p = p\left(n, m\right) = \left(1 - e^{-\frac{m}{n} \cdot \ln(2) \cdot \frac{n}{m}}\right)^{\frac{m}{n} \cdot \ln(2)} = \left(1 - e^{-\ln(2)}\right)^{\frac{m}{n} \cdot \ln(2)} = \left(\frac{1}{2}\right)^{\frac{m}{n} \cdot \ln(2)}$$

Taking the base 2 logarithm of both sides:

$$\log_2\left(p\right) = \log_2\left[\left(\frac{1}{2}\right)^{\frac{m}{n} \cdot \ln(2)}\right] = -\frac{m}{n} \cdot \ln\left(2\right)$$

and then solving for `m` finally gives us:

$$m^* = -n \cdot \frac{\log_2\left(p\right)}{\ln\left(2\right)} = -n \cdot \frac{\ln\left(p\right)}{\ln\left(2\right)^2}$$

This means that, if we know in advance the total number `n` of unique elements that we will insert in the Bloom filter, and we set the false positive probability `p` to the largest value that is acceptable, then we can compute the optimal size of the buffer that guarantees the desired precision; we will also have to set `k` accordingly, using the formula we have derived for it:

$$k^* = \ln\left(2\right) \cdot \frac{m^*}{n} = -\ln\left(2\right) \cdot n \cdot \frac{\ln\left(p\right)}{\ln\left(2\right)^2} \cdot \frac{1}{n} = -\frac{\ln\left(p\right)}{\ln\left(2\right)}$$

For instance, if we would like to have a precision of 90%, and so at most 10% false positives, we can plug in the numbers and get: